# Reading the Voids

Spherical waves, harmonics and golden bricks find, describe and fill the gaps of a packing; the Dehn invariant says exactly what can never be filled



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Paper #96 · version 3 · 5 November 2026 · DICTO DOI: 10.5281/zenodo.23968272 (concept) · Licence CC BY-SA 4.0 · Data, scripts and checks: https://github.com/DICTOR-Master/kaleidohedra (branch research, studies 2026-11-05-spherical-waves, 2026-11-05-golden-skin, 2026-11-05-og-pentagons)
DICTO proposed spherical wave patterns to reveal and close gaps, used together with harmonics; reading bonds between atoms and views inside cells; growing golden or regular rhombi in the gaps; the original Ogdoecontahedron with real rhombi for a golden solution; and his Bull-horns adaptor as the basis. Computation, proofs and drafting with Claude (Anthropic).

A packing of solids leaves gaps. Most of the time we only measure how much. Here we look at them: we send waves through the packing and the gaps show up dark; we read the dark places like a chemist reads the cloud of electrons in a crystal, as rooms joined by windows; we listen to each room's shape; and we grow golden bricks into it until it is full. One thing no brick can ever fill, and an old invariant from 1901 tells us exactly what it is.

**Abstract.** Grown from DICTO's work on harmonics and resonance, where single solids were heard as tones, we let spherical waves run through whole packings and assemble a toolkit for their gaps and try it on DICTO's Ogdoecontahedron (OG), its pentagon-cut form (the cut OG), and DICTO's Bull-horns OG adaptor. **Theorem 1** (exact): the OG and the cut OG have nonzero Dehn invariants; they share their 120 hexagon-rim edges, and the obstruction lives only on the edges. **Theorem 2** (exact): the room adaptor of every lattice packing of either solid, and DICTO's Bull-horns OG adaptor, carry a nonzero Dehn invariant (minus the OG's, or a third of it), so none of them can be cut into golden rhombohedra or any pieces whose dihedral angles are rational multiples of π. Then the tools (float): spherical waves find the gaps of the densest lattice packing of the cut OG, never mapped before; reading the wave field by its critical points gives, per copy, one atom, five bonds, five rings and one cage: one gap room opening through eight windows; the room closes with one piece, the DICTO Void Keeper, so the cut OG and the Void Keeper fill space by translation. Golden rhombohedra grown in φ-steps fill 84% of the room; one coherent golden tiling completes it with whole and trimmed bricks; around the original OG a complete fill by the OG, the DICTO Golden Skin and one golden rhombohedron makes the skin as thin as wished (2.9% of space), and the Bull-horns adaptor is completed with 66% whole golden bricks. Golden tetrahedra, whose Dehn invariants are not zero, cannot rescue a completely golden fill either: their angles are of another kind than the OG's (strong numerical evidence).

## Where this began: tones, resonance and waves

The toolkit grew out of listening. In The Silent Golden Station (#94) a solid's corners, seen from its centre, became notes on a sphere, each degree of spherical harmonic a tone: on the sweep from the octahedron to the cube the cubic tone falls silent exactly once, at Icosaxes. In four dimensions the same test hears the 600-cell, DICTO's White Queen, as silent through

every tone up to degree 11. DICTO then widened the search to harmonics, music, waves and particles and resonance (The DICTO Concordance, Part 5): in The Golden Ladder the golden stations of the sweep became the limits of ladders of particle multiplets, half the sweep one musical octave, and at its top rung the rings of the E8 picture stand in the ratios of the eight masses that resonate in a chain of magnetic spins, the second φ times the first. Each of these heard one shape at a time. The next step, DICTO's, was to let waves run through a whole packing in three dimensions, where they meet, add and cancel between the pieces: spherical waves. Where they cancel, the gaps are.

## 1. The toolkit

**Spherical waves.** Every point of the solid sends out a spherical wave sin(kr)/(kr), the kernel of Debye's scattering formula (Debye 1915); the waves add up to a field over all space, computed by fast Fourier transform (Cooley and Tukey 1965) on one period. Where they cancel the field is dark. At the packing's own wavelength the dark places are the gaps. **Critical points.** As in the electron density of a crystal (Bader 1990), the field's peaks are atoms, its saddles with two falling directions are bonds, those with one are rings (windows) and its minima are cages (rooms); in a periodic field atoms − bonds + rings − cages = 0 (Morse theory: Milnor 1963), a built-in check. **Harmonics.** The shape of a room heard from its centre, as in DICTO's harmonic fingerprint (#94): odd tones vanish for a room with a centre of symmetry, the l = 4 tone is cubic, icosahedral shapes have none. **Golden bricks.** Golden rhombohedra with edges along the solid's own five-fold axes, grown greedily, darkest places first, in the manner of random sequential adsorption (Feder 1980), or cut from one icosahedral tiling of golden rhombohedra (de Bruijn 1981; Kramer and Neri 1984; Socolar and Steinhardt 1986; Katz and Duneau 1986), in φ-steps from coarse to fine. **Exact certificates.** Every kept brick lies wholly inside its room and every pair of bricks is disjoint, checked one by one with solid booleans (the Manifold library, Lalish et al.); the rest of the room is the finest bricks trimmed by the walls, which cover it because a tiling covers space. **The Dehn invariant** decides what bricks can never do (§2).

## 2. What no brick can fill

**Theorem 1.** The OG and the cut OG have nonzero Dehn invariants.

**Proof.** In exact coordinates ($\mathbb{Q}(\sqrt{5})$) each solid has two kinds of edge. OG: 60 kite–kite edges of length $(9 + \sqrt{5})/57$ at dihedral angle 156.7186°, and 120 kite–hexagon edges of length $\sqrt{(2/9 - 2\sqrt{5}/27)}$ at 156.1998°. Cut OG: 60 triangle–pentagon edges of length 2/9 at 159.9232°, and the same 120 hexagon-rim edges at 156.1998°. In each solid the two lengths are rationally independent (the rim length involves $\sqrt{3}$, or its ratio to 2/9 is not a square in $\mathbb{Q}(\sqrt{5})$), so the invariant vanishes only if both angles are rational multiples of π. Each cosine has degree at most 4, so an angle kπ/n would need $\varphi(n) \le 8$, $n \le 30$; no such angle matches. ∎

**Theorem 2.** Let A be the room adaptor of a lattice packing of the OG or the cut OG (the Voronoi cell of the room centres minus the solids), or DICTO's Bull-horns OG adaptor in the body-centred double network. Then Dehn(A) = −Dehn(OG), −Dehn(cut OG), or −Dehn(OG)/3 respectively, never zero. No dissection of A uses only golden rhombohedra, rhombic-dodecahedron rhombohedra, cubes, or any pieces whose dihedral angles are rational multiples of π.

**Proof.** Lattice case: the Voronoi cell C of a lattice is a parallelohedron, a zonohedron, with Dehn(C) = 0. C is A together with the parts of the solids that fall in C; translated by lattice vectors these parts reassemble exactly one solid, and the Dehn invariant adds over dissections, so 0 = Dehn(A) + Dehn(solid). Bull-horns: a cube of the double network's period holds, after translating parts back, two OGs and six adaptors, all congruent, so 0 = 2·Dehn(OG) + 6·Dehn(A). By Theorem 1 these are nonzero, and pieces with rational dihedral angles all have zero Dehn invariant. ∎

So every golden filling below must leave pieces that are not golden rhombohedra; the question is only how few and how thin.

## 3. Waves reveal the gaps

**A known case first.** On the diamond network of regular-triangle cut OGs (the pyrochlore network of Pentagons for the Kites, #95), at the network's own wavelength the darkest half of space lies wholly in the gaps: it covers 80% of them and none of the solid (Figure 1).

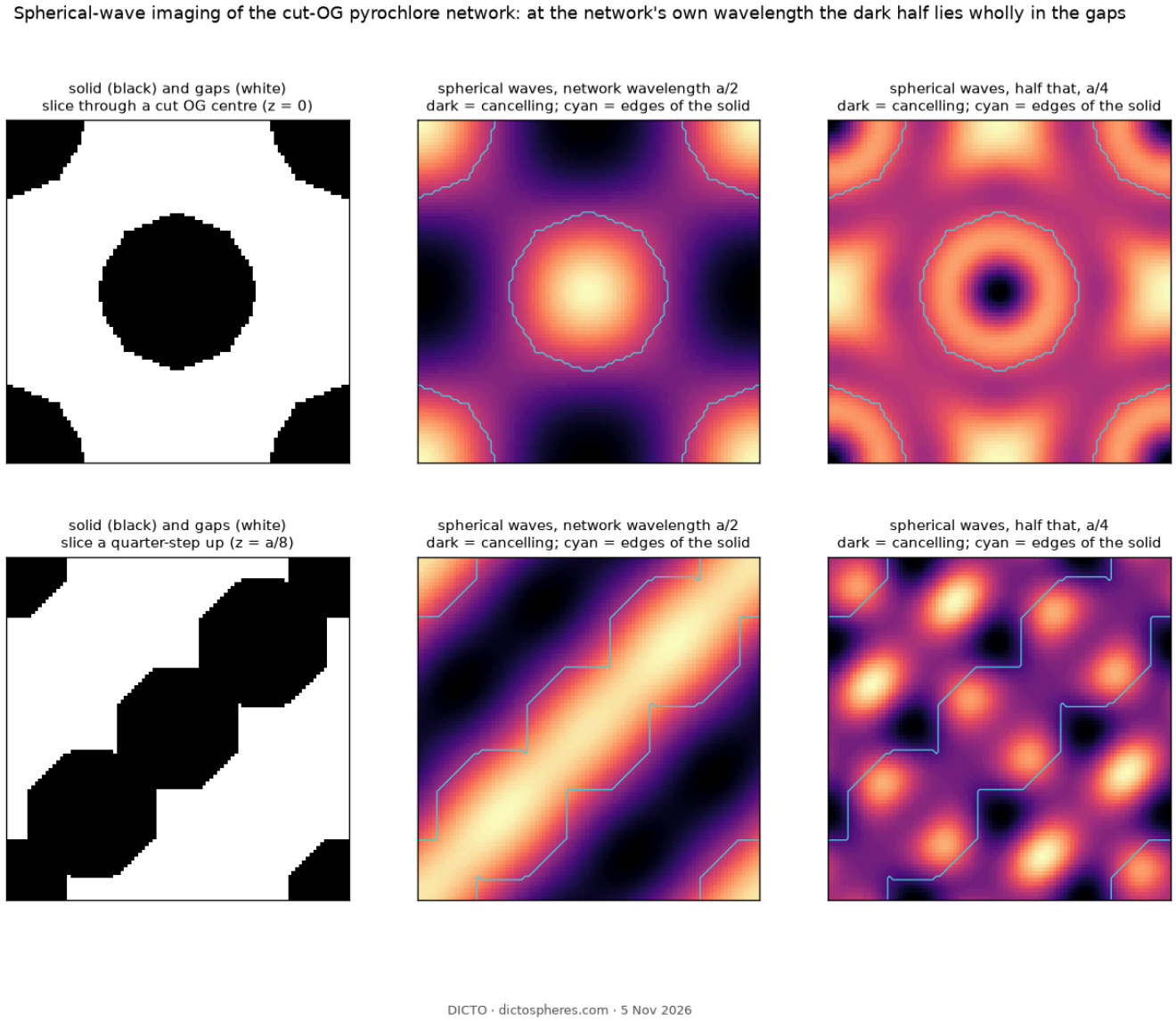



Figure 1. The cut-OG pyrochlore network: the solid (left) and the spherical-wave field at the network's own wavelength and at half of it. At the right wavelength the dark places are the gaps.

**An unknown case.** The densest lattice packing of the cut OG (density 0.776244) had never been looked at. Its field has, per copy, **one atom, five bonds, five rings and one cage** (1 − 5 + 5 − 1 = 0; grids 48 and 72 agree). The cage is one gap room per copy, at half a lattice vector, a centre of symmetry of the packing. Four ring points per cell lie in the gap, so each room opens through **eight windows**: the gap is one connected network of rooms. Of the twelve contacts of a copy, five pairs read as bonds and one pair as a ring point inside the solids: a weak contact (Figure 2).

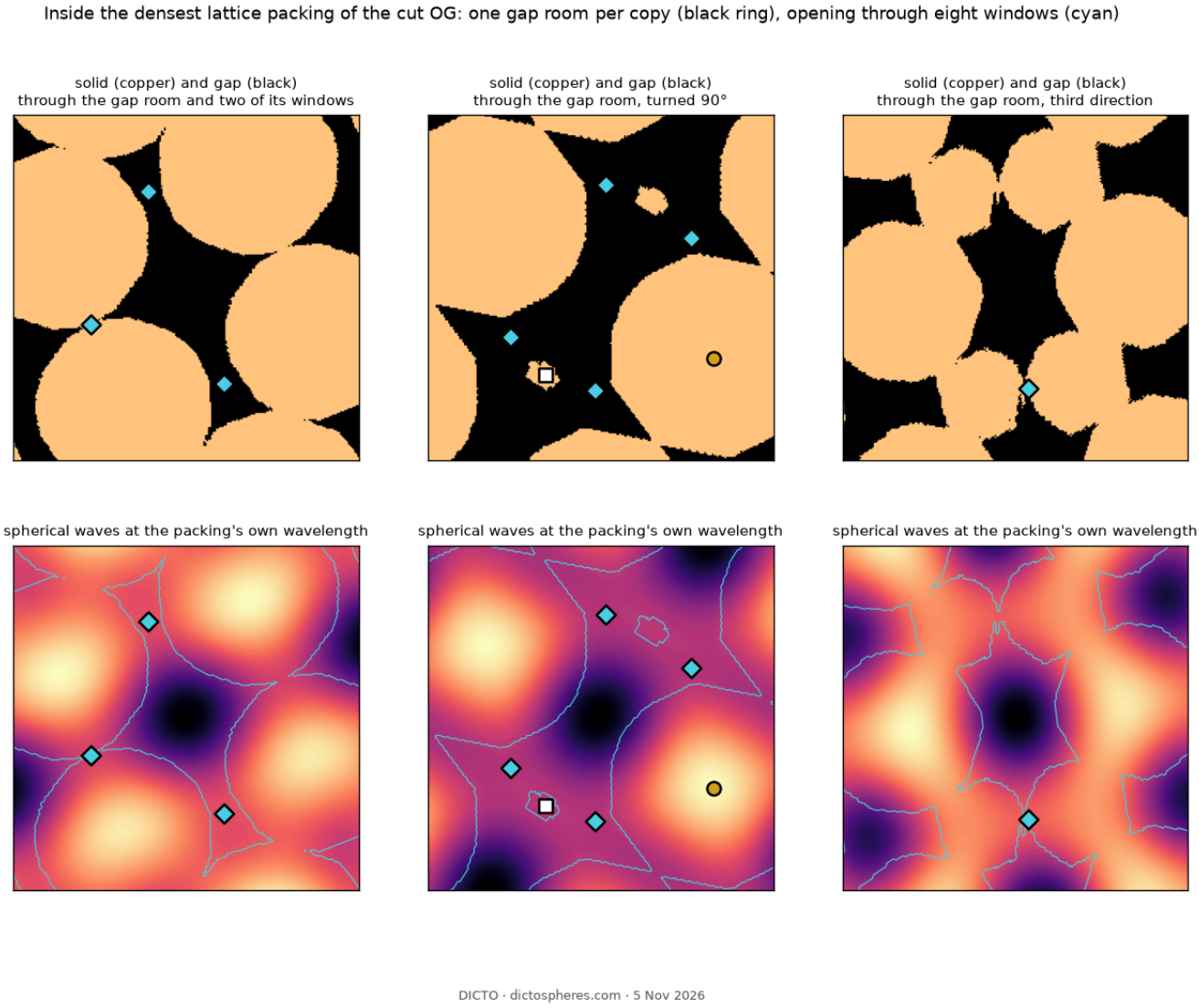



Figure 2. Inside the densest lattice packing of the cut OG: three planes through the gap room. Top: solid and gap. Bottom: the wave field, dark in the room; windows cyan.

**Closing the room.** The Voronoi cell (Voronoi 1908; Fedorov 1885) of the room centres minus the cut OGs is one solid without holes, of volume 0.326555, exactly the gap per copy; copies overlap neither each other nor the cut OGs. The cut OG (77.6%) and this room adaptor, named by DICTO the **DICTO Void Keeper** (D133; 22.4%), fill space, both by translation only (Figure 3). Together they are **DICTO Void set 1 (cut OG)**: piece V1, the cut OG, and piece V2, the Void Keeper. It fills 31% of its convex hull, sphericity 0.374.

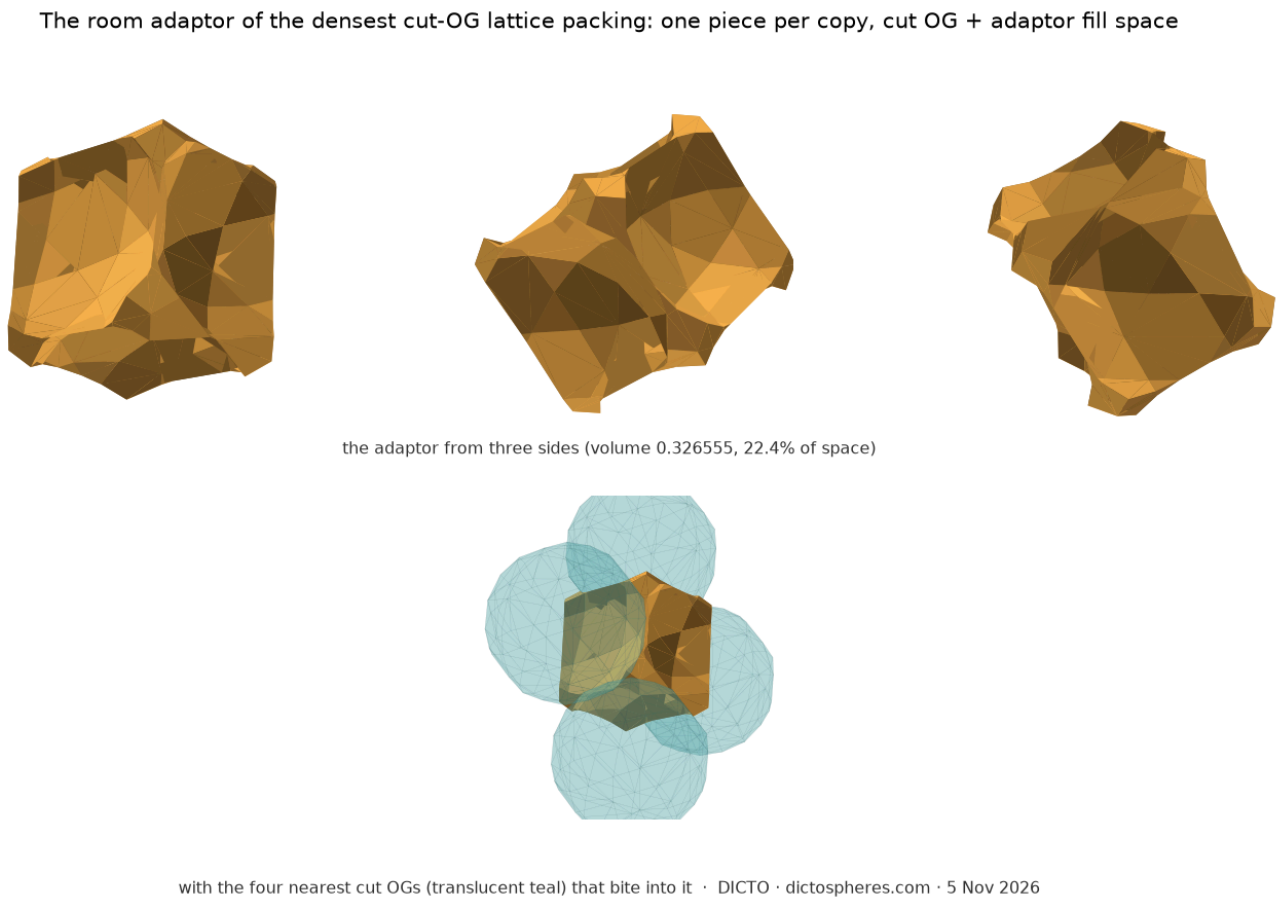



Figure 3. The DICTO Void Keeper, the room adaptor of the densest cut-OG lattice packing, from three sides, and with the four nearest cut OGs that bite it.

## 4. Golden bricks fill them

**Harmonics choose, walls decide.** The room's harmonics have no odd tones and half their weight in l = 4, a cubic tone; yet golden rhombohedra fill it better than rhombic-dodecahedron rhombohedra (39% against 37% at one size), because they share the cut OG's five-fold directions. Grown in φ-steps, darkest places first, golden bricks fill 27, 42, 58, 78 and 84% of the room; adding the regular bricks gains almost nothing. What is left is one connected web: a skin on the walls (42% of it within one grid step of the solid, against 19% of all gap) and seams where bricks grown from different seeds meet.

**One coherent grain completes the room.** Bricks cut from one icosahedral tiling meet without seams. Kept only when wholly inside the room and clear of all others, 1055 whole golden rhombohedra in three φ-steps fill 38% of the room; the finest bricks of the same tiling, trimmed by the walls, fill the rest: whole + trimmed = the room to $7 \times 10^{-5}$ (float). By Theorem 2 the trimmed pieces can never disappear.

**The original OG in a golden skin.** Put the OGs on a lattice spanned by three of their own five-fold axes (a scaled prolate golden rhombohedral lattice, OG density 0.5633) and tile space by the prolate golden rhombohedron of edge t/m. The bricks that touch an OG, minus the OG, form its skin, named by DICTO the **DICTO Golden Skin** (D134, one for each brick size). OG + Golden Skin + whole bricks fill space exactly, at every m, with three shapes: **DICTO Void set 2 (OG)**, pieces V3 the OG, V4 the Golden Skin and V5 the prolate golden rhombohedron. The skin halves when the brick edge halves (32.3, 22.1, 11.5, 5.9, 2.9% of

space at m = 8, 12, 24, 48, 96): at m = 96, 93% of the gap is whole golden bricks (Figure 4). By Theorem 1 the skin is never zero.

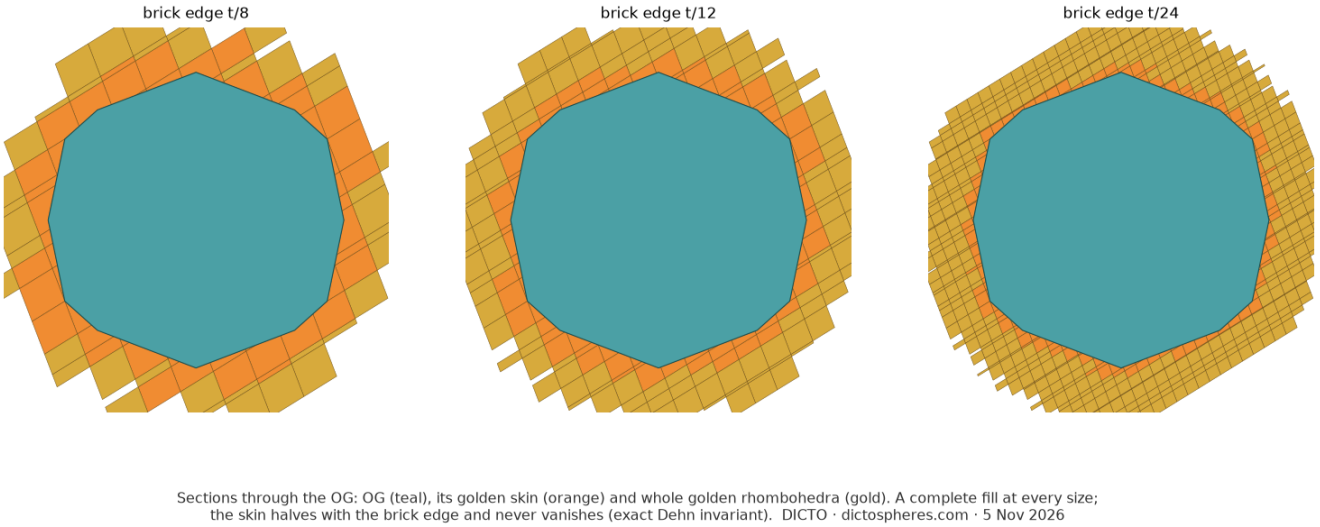



Figure 4. Sections through the original OG (teal), its DICTO Golden Skin (orange) and whole golden rhombohedra (gold) at three brick sizes: a complete fill each time, the skin thinning.

## 5. DICTO's Bull-horns adaptor, made golden

In the body-centred double network the OGs fill 77.4% of space and DICTO's Bull-horns adaptor the rest, one shape, three per OG (#36). Rebuilt from its definition (volume 0.111898), the adaptor is completed by one coherent golden tiling: whole golden rhombohedra fill 25, 37, 49, 58 and 66% of it in five φ-steps (12483 bricks), and the trimmed finest bricks the rest; whole + trimmed = the adaptor to $3 \times 10^{-5}$ (Figure 5). The bricks fill the body; the thin horns that reach between the OGs are where the trimmed pieces sit.

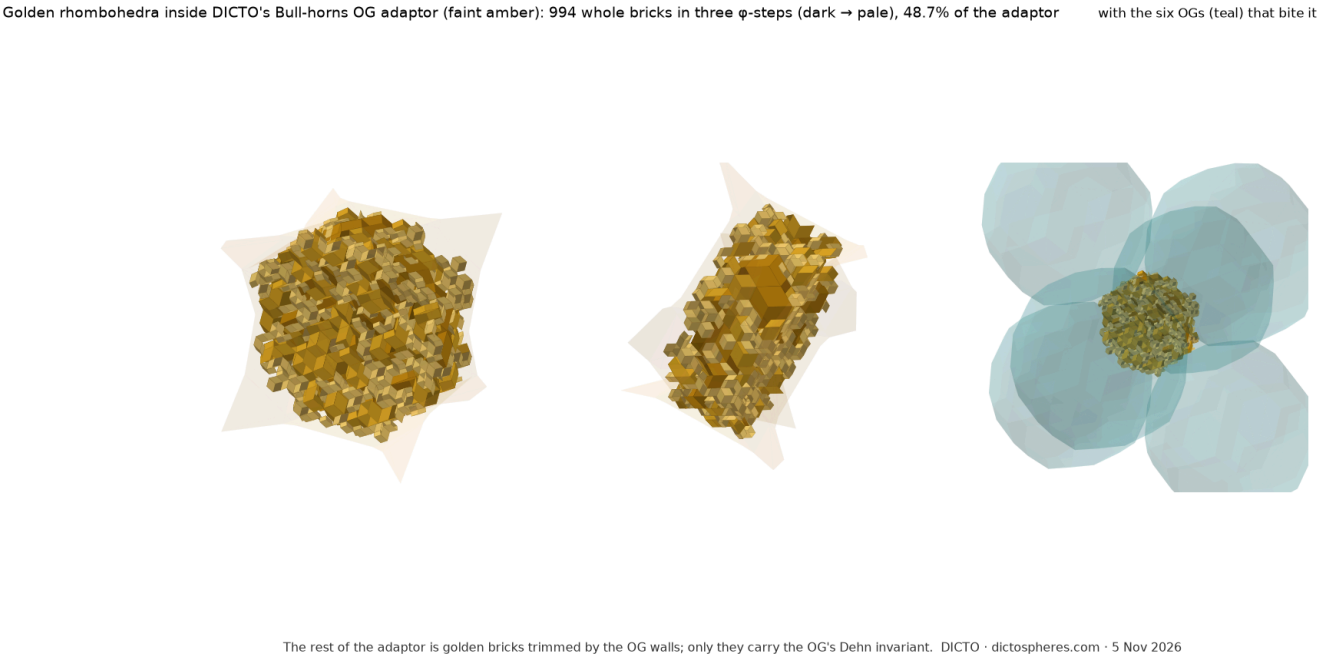



Figure 5. Golden rhombohedra inside DICTO's Bull-horns OG adaptor (faint amber), three φ-steps, and with the six OGs that bite it.

## 6. No completely golden fill

Golden rhombohedra cannot do it (Theorem 2). Golden **tetrahedra** do carry a Dehn invariant: Ogievetsky and Papadopolos (1999) show that every dihedral angle of the six golden tetrahedra is, modulo π, a rational combination of two angles, $\beta = \arccos((\tau+1)/(\sqrt{3}\cdot\sqrt{(\tau+2)}))$ and $\delta = \arccos((\tau-1)/(\sqrt{3}\cdot\sqrt{(\tau+2)}))$, and the Mosseri–Sadoc tiles, among them DICTO's AXE and CLEO (#19), lie in the same span. Golden pieces of any scale can therefore cancel a Dehn invariant only if its angles are rational combinations of β, δ and π. A search for integer relations (PSLQ at 80 digits, coefficients up to a million; their relations $\alpha + \gamma + 2\beta = \pi$ and $\alpha - \gamma + 2\delta = \pi$ recovered to 81 digits) finds none between β, δ, π and any of the OG's angles (hexagon rim 156.1998°, kite ridge 156.7186°) or the cut OG's pentagon edge (159.9232°), singly or together. So, on strong numerical evidence, no fill by golden pieces alone can close around the OG: the DICTO Golden Skin is as golden as it gets. A proof would need an independence result for angles whose squared cosines lie in $\mathbb{Q}(\sqrt{5})$, in the spirit of Conway, Radin and Sadun. Still open: exact forms of the Void Keeper and the Golden Skin; splitting the skin into a few repeating edge wedges; a denser golden lattice for the OGs.

## Known and new

**Known:** the Dehn invariant and its additivity (Dehn 1901; Boltianskii 1978); zonohedra have zero Dehn invariant; critical points of densities (Bader 1990) and their count (Milnor 1963); the scattering kernel sin(kr)/(kr) (Debye 1915) and the fast Fourier transform (Cooley and Tukey 1965); Voronoi cells and parallelohedra (Voronoi 1908; Fedorov 1885); random sequential adsorption (Feder 1980); spherical-harmonic order parameters (Steinhardt, Nelson and Ronchetti 1983); the dual-grid and cut-and-project constructions of icosahedral tilings by golden rhombohedra (de Bruijn 1981; Kramer and Neri 1984; Socolar and Steinhardt 1986; Katz and Duneau 1986); Dehn invariants in icosahedral tilings (Ogievetsky and Papadopolos 1999). **New here, as far as the sources checked go:** Theorems 1 and 2 for the OG, the cut OG and their adaptors; the gap reading of the densest cut-OG lattice packing and the DICTO Void Keeper; the golden fills and the DICTO Golden Skin. The tools are classical; their use together on a sculptor's solids is this paper's contribution.

**Sources checked:** the reference library (searches “Steinhardt bond orientational order; Bader critical points; de Bruijn dual grid; Kramer Neri; Dehn invariant zonohedra”, “E8 icosians 600-cell”, 5 Nov 2026), Ogievetsky–Papadopolos 1999 read (abstract and introduction); for Version 2 the method sources were added to the library (link entries: Debye 1915, Milnor 1963, Voronoi 1908, Feder 1980, Socolar–Steinhardt 1986, Katz–Duneau 1986, Cooley–Tukey 1965, Manifold); Fedorov 1885 and Dehn 1901 are held in it.

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